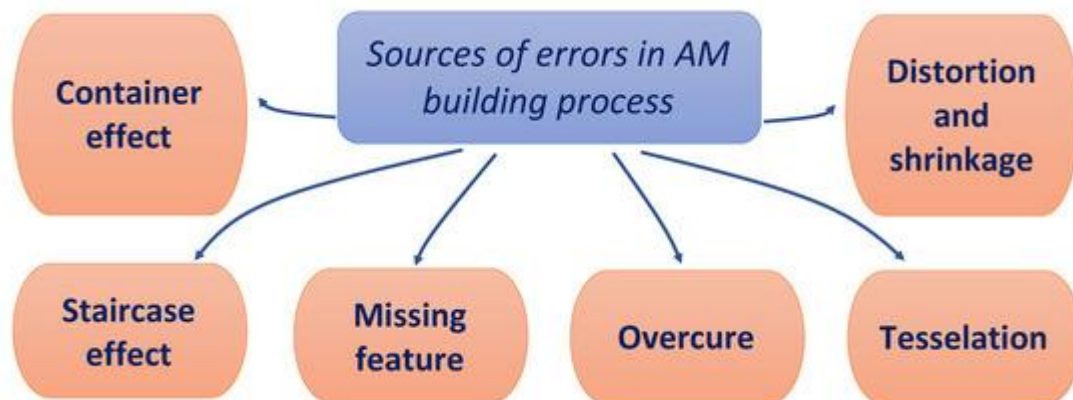


### ***RAPID MANUFACTURING PROCESS OPTIMIZATION***

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#### **COURSE CONTENT**

- 3.3.1 Introduction**
- 3.3.2 Errors during Data Preparation**
- 3.3.3 Errors during the Part building**
- 3.3.4 Errors during Part finishing**

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### 3.3.1 INTRODUCTION

Accuracy in 3D printing refers to the degree to which the printed part's dimensions and geometry match the original CAD model. It is commonly measured in terms of dimensional deviation or tolerance, such as  $\pm 0.1$  mm or  $\pm 0.2\%$  of the part size. Achieving high accuracy is essential for ensuring proper fit, function, and overall performance of additively manufactured components.

Optimizing the AM process plays a key role in improving dimensional accuracy, surface quality, and the mechanical performance of the final part. In AM, several factors influence overall process accuracy. These factors are generally classified into three major categories:

1. Errors during Data preparation: includes
  - STL file generation
  - Model slicing
  - Part build direction
2. Errors during the Part building:
  - Process-specific parameters such as layer thickness, scan speed, curing or melting behaviour and machine calibration etc.
3. Errors during Part finishing:
  - Surface finishing, cleaning, curing, and other secondary operations

Understanding and controlling these factors is crucial for producing parts that closely replicate the intended CAD design and meet the required functional specifications.

### 3.3.2 ERRORS DURING DATA PREPARATION

#### 1. STL FILE ERRORS:

Errors caused by tessellation are often overlooked due to the common assumption that increasing the number of triangular facets will automatically minimize these errors. However, in practice, the number of triangles cannot be increased indefinitely due to limitations in file size, and processing speed.

Therefore, the resolution of STL files must be carefully controlled during their generation in a 3D CAD system by adjusting the tessellation parameters. During STL generation, two primary parameters influence accuracy:

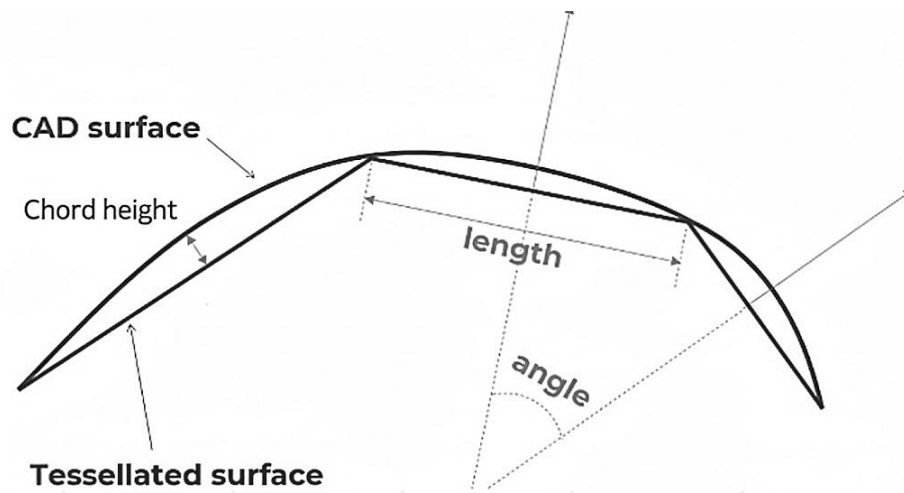
#### a. Chord Height:

- Chord height specifies the maximum allowable distance between the actual curved surface and the straight chord connecting two points on that surface.
- A smaller chord height reduces deviation from the actual geometry, resulting in higher accuracy.
- The upper limit of chord height depends on the overall model size, while the lower limit is governed by the accuracy of the CAD model itself.

#### b. Angle Control:

- Angle control defines the maximum permitted angle between the normals of adjacent triangular facets.
- A low angular tolerance results in high accuracy and smoother surfaces but produces a larger STL file size.

- In contrast, a high chord height combined with a high angular tolerance leads to lower accuracy and more faceted surfaces but generates a smaller file size.



***Tessellation Error in CAD***

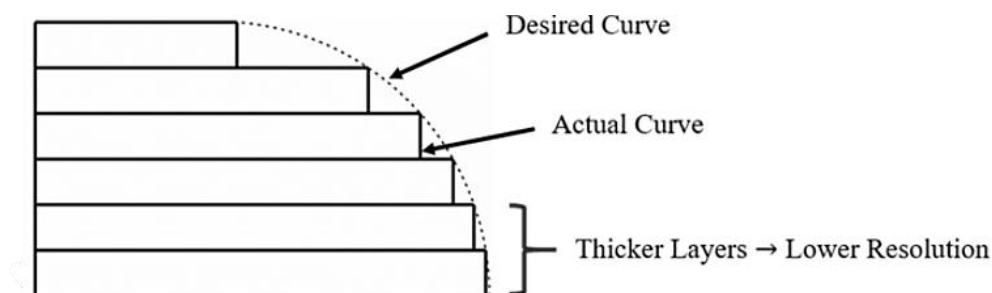
Reducing chord height and tightening angular tolerance improves geometric fidelity and produces smoother surfaces. However, this also increases the number of facets, resulting in significantly larger STL files and higher processing requirements. Therefore, it is essential to choose parameter values that balance accuracy with file size.

## 2. SLICING ERRORS:

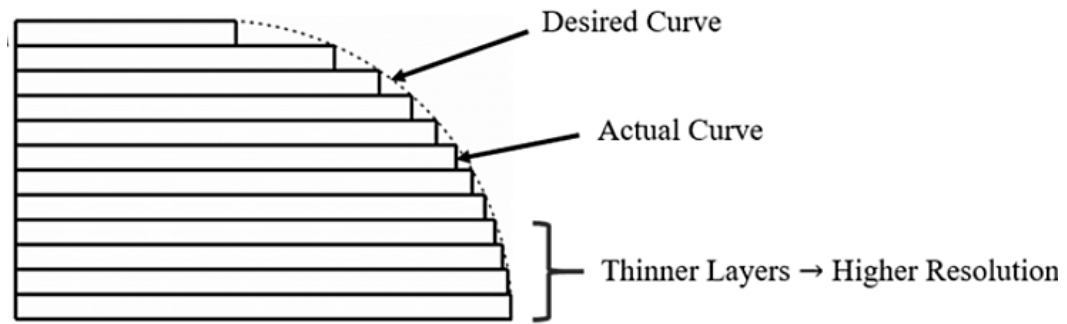
In AM, slicing is a critical step that significantly influences the dimensional and surface accuracy of the final part. AM systems fabricate components layer by layer, and this discrete layering inherently introduces the stair-stepping effect, particularly on curved or inclined surfaces. Because each layer has a finite thickness, the original CAD geometry cannot be reproduced continuously, instead it is approximated by step-like edges. This approximation leads to deviations between the desired curve and the actual printed contour, affecting overall part accuracy.

The magnitude of this error is strongly dependent on layer thickness:

- Thicker layers result in larger step heights, producing lower surface resolution and reduced dimensional accuracy.
- Thinner layers more closely follow the curvature of the CAD model, enhancing surface resolution and improving the accuracy of the printed part though at the cost of increased build time.



***Thick layers - reduced dimensional accuracy***

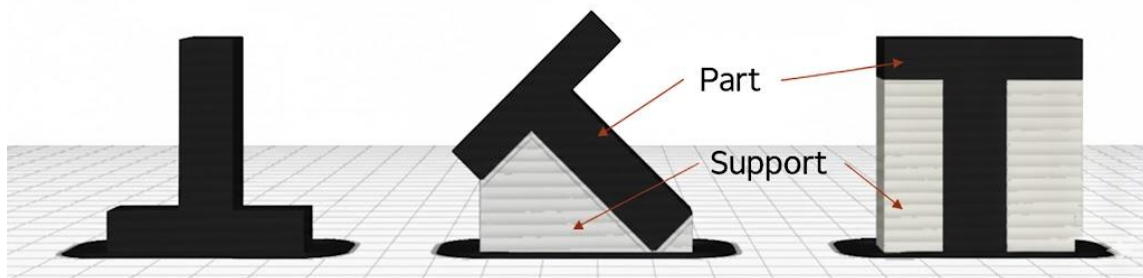


***Thin layers - improved dimensional accuracy***

Curved profiles in a CAD model are mathematically defined by their curvature and radius of curvature. For accuracy analysis, especially in Additive Manufacturing, these curves are often approximated using arc segments. Arcs provide simpler mathematical handling, constant curvature, easier error estimation, and better control over tessellation and slicing parameters. This makes them highly suitable for evaluating stair-stepping effects and understanding how closely the printed layers follow the intended geometry.

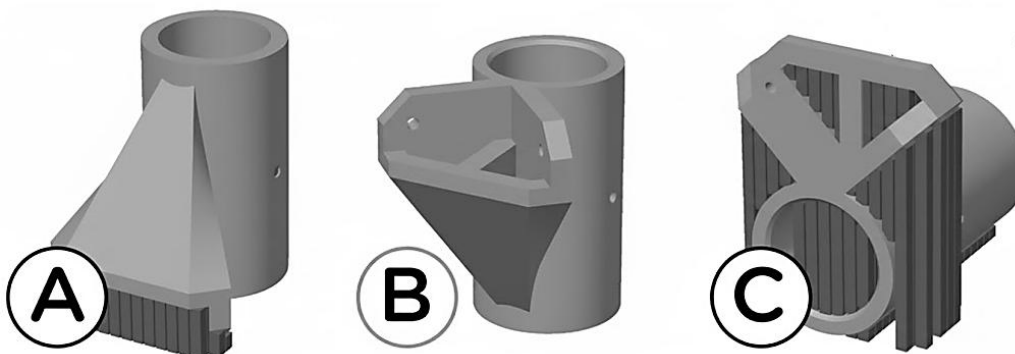
### 3. PART BUILDING ORIENTATION:

The selection of part build orientation is one of the most critical decisions in AM, as it directly influences accuracy, mechanical performance, build time, and overall manufacturing cost. Proper orientation helps minimize dimensional errors, reduce surface defects, and achieve better structural integrity. Since AM parts generally exhibit weaker mechanical properties along the Z-axis, orienting the part to place major loads within the XY plane significantly enhances strength.

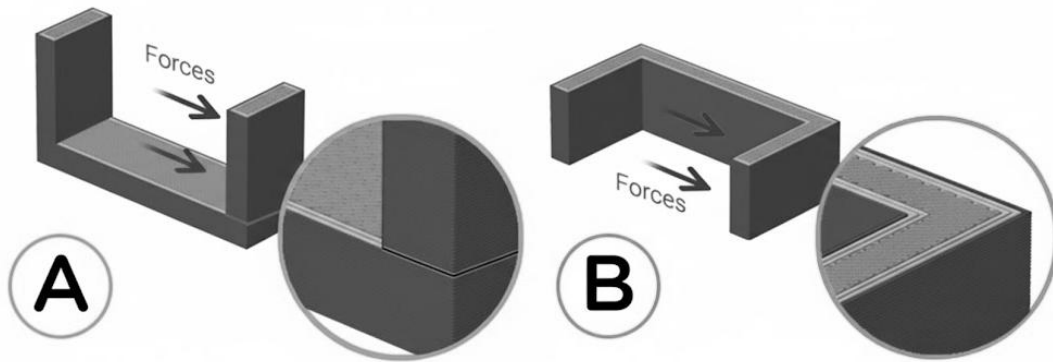


***Part build orientation***

**Example 1:** The part in Orientation B is ideal because it minimizes or eliminates the need for support structures, unlike orientations A and C, which require excessive supports.



**Example 2:** The part should also be printed in orientation B, where the applied forces act across the layers, resulting in stronger walls. Printing in orientation A makes the part weak because forces align with the layer lines, causing the walls to snap off easily.



An optimal build orientation should aim to:

- Minimize the height of support structures, reducing material usage and post-processing effort.
- Reduce the overall height of the part, thereby decreasing the number of layers and shortening build time.
- Limit inclined surfaces, which are more prone to stair-stepping errors and poor surface finish.
- Align critical load-bearing directions parallel to the layer plane ( $XY$ ) to improve mechanical performance and durability.

Guidelines for Feature Orientation in AM:

- Holes are oriented orthogonally to the horizontal planes to prevent difficult support removal and to minimize stair-stepping on the internal surfaces.
- Cuts that pass entirely through the part or exceed a minimum depth are aligned such that the intersecting planes become horizontal; shallow or partial cuts are generally ignored to avoid unnecessary complexity.
- Shaft features are oriented so that their defining planes lie horizontally, ensuring improved external surface finish and reduced stair-stepping.
- Protrusions formed by revolving a profile are positioned with their axes vertically aligned to achieve better accuracy and reduce surface imperfections.
- Shell features are oriented with the concave surface facing upward to reduce the need for internal supports and improve the quality of inner surfaces.
- All axes within the model are aligned vertically wherever possible to enhance build stability, accuracy, and surface finish.

### 3.3.3 ERRORS DURING THE PART BUILDING

(a) PART BUILDING ERRORS IN SLA:

In SLA, part-building accuracy is influenced by errors occurring during the curing and scanning stages. These errors affect dimensional precision, surface finish, and mechanical strength. The major types of SLA part-building errors include:

1. Curing Errors: Curing errors occur due to improper laser exposure during the photopolymerization process.

- Overcuring: A controlled amount of overcuring is necessary to ensure proper bonding between layers and achieve fully solid parts. However, excessive laser exposure causes the resin to cure beyond the intended boundaries. It can lead to Dimensional inaccuracies, Positional errors and loss of fine details and sharp edges.
- Undercuring: Occurs when laser exposure is insufficient to fully cure the resin in each layer. The resin remains partially polymerized, reducing overall strength and layer adhesion. Undercuring can cause Weak or brittle parts, delamination between layers and distorted geometry

2. Control Errors: Control errors are associated with layer thickness selection and scan positioning accuracy.

- Layer Thickness: Using large layer thickness helps reduce build time, but it leads to noticeable stair-stepping on curved or inclined surfaces, poor surface finish that requires more post-processing and weaker interlayer bonding, which lowers mechanical strength. In contrast, using small layer thickness improves accuracy but increases the overall build time, makes the process more sensitive to exposure variations that may result in overcuring and risk of print failure.
- Scan Position Control: Accurate scan positioning ensures that each point is cured exactly at its designated location. Any deviation or misalignment introduces positional errors, geometric distortions, and dimensional deviation in the part. Poor scan control may also lead to stitching errors between scan paths.

**(b) PART BUILDING ERRORS IN FDM:**

1. Warping: Warping occurs when printed layers cool unevenly, causing thermal contraction that lifts the corners of the part from the build plate. It is common in high-shrink materials like ABS and Nylon and results in dimensional inaccuracies. This defect can be minimized by using a heated bed, enclosing the printer, adding brims or rafts and reducing fan cooling during the initial layers.

2. Layer Shift: Layer shift happens when an entire layer is printed in the wrong position due to unintended movement of the print head or build platform. It is usually caused by loose belts or pulleys, stepper motor skipping, excessive vibrations. The result is misaligned geometry and distorted features, which can be prevented by tightening mechanical components, reducing print speed and lubricating rails.

3. Poor Interlayer Bonding: Poor interlayer bonding occurs when successive layers do not fuse properly, producing weak prints. This defect is typically caused by insufficient nozzle temperature, excessive print speed and moisture-laden filament. It leads to cracks between layers and reduced strength along the Z-axis and can be prevented by using the correct print temperature, slowing down printing and storing filament in a dry environment.

4. Surface Roughness: Surface roughness can be caused by large layer height, inconsistent extrusion flow, worn nozzles, and improper print speed settings. This defect affects dimensional accuracy and aesthetics, but can be improved by selecting smaller layer heights, maintaining the extrusion system, and optimizing speed and temperature settings.

5. Nozzle Clogging: Nozzle clogging happens when the nozzle becomes partially or fully blocked, preventing smooth extrusion of filament. It is caused by dust, degraded filament, moisture in the filament and incorrect temperature settings. This leads to under-extrusion, missing layers, and possible print failure, and can be prevented by using clean and dry filament, maintaining proper temperature, and regularly cleaning or replacing the nozzle.

(c) PART BUILDING ERRORS IN SLS:

1. Powder Bed Defects: Powder bed defects occur when the powder is not evenly distributed across the build platform. Uneven powder spreading leads to variations in layer thickness, poor surface quality and porosity. Common causes include worn recoater blades, contaminated powder, moisture, and improper powder flowability.

2. Layer Irregularities: Layer irregularities arise when each deposited powder layer is not uniform in thickness. This leads to inconsistent energy absorption during laser scanning, causing distorted geometry, reduced dimensional accuracy, and weak zones within the part.

3. Thermal Distortion: Because SLS involves high temperatures, uneven heating and cooling can cause warping or curling of the part during building. Differences in thermal expansion can create internal stresses that distort the geometry or cause layer separation.

4. Incomplete Sintering: If the laser power or scan speed is not properly controlled, the powder may not fully fuse, leading to weak bonding between particles. This results in fragile parts, poor mechanical strength, and grainy or crumbly surfaces.

5. Over-sintering: Excessive laser exposure causes the powder to melt beyond the intended boundaries. This leads to dimensional inaccuracy, loss of fine features, surface roughness, and possible fuse-together of moving parts designed with clearance.

### **3.3.4 ERRORS DURING PART FINISHING**

Part finishing is an essential stage in AM, especially for producing high-quality models and master patterns used in indirect tooling. Part finishing is a crucial phase in AM, particularly for components intended for high-quality visual models, functional prototypes, or master patterns used in indirect tooling. Finishing operations enhance both the surface appearance and dimensional accuracy of the printed part.

A variety of processes can be used to finish AM models. Common finishing methods include wet and dry sanding, sand blasting, coating, spray painting, infiltration with specialized solutions, machining, and several others.

Finishing techniques that offer better dimensional control tend to have less impact on the accuracy of the final model. For example, machining or milling-based finishing produces more precise results and causes minimal dimensional deviation, whereas manual wet sanding or sand blasting can remove more material unevenly, thereby having a greater influence on the model's overall accuracy.

Key sources of finishing-related errors include:

- Stair-Stepping Removal: Curved and sloped surfaces exhibit visible stair-stepping due to layer-by-layer fabrication. Removing these steps through sanding, polishing, bead blasting, or vapor smoothing is essential to achieve a smooth and accurate surface.

- Support Removal Damage: Poor support design or excessive support structures can result in surface marks and dents during removal.
- Dimensional variation: Techniques such as sanding or grinding can unintentionally remove excess material, leading to loss of dimensional accuracy. Over-polishing can distort edges, radii, and fine features.
- Thermal & Chemical finishing Issues: Chemical treatments may cause over-melting, softening edges or altering tolerances. Heat-based smoothing can lead to warping or uneven gloss levels.
- Incomplete Cleaning: Powder-based AM processes may leave embedded powder or un sintered particles, affecting surface quality and coating adhesion.